

上海交通大学
巴黎卓越工程师学院
中国上海

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研究概述

My undergraduate research focused on computer vision, with an emphasis on visual recognition and computational photography. During my master's studies, my research has centered on embodied intelligence, including zero-shot navigation, robotic manipulation, and post-training for vision-language-action models. My current work investigates the integration of world models and navigation, with particular emphasis on the relationship between learned world-model representations and navigational maps. In doctoral study, I aim to further develop world models and embodied agent systems for robust perception, planning, and interaction in physical environments.

研究方向： 具身策略与智能体系统，世界模型与视觉基础模型，强化学习算法与系统。

教育经历

M.Eng. Sep 2024–Mar 2027 (预计)	上海交通大学 (SJTU) SJTU Paris Elite Institute of Technology (SPEIT) 电子信息 中国上海.
交换生 Sep 2024–Feb 2025	里昂国立应用科学学院 (INSA Lyon) 计算机科学与信息技术系 (IF) 法国里昂.
B.Eng. Sep 2020–Jun 2024	上海交通大学 (SJTU) SJTU Paris Elite Institute of Technology (SPEIT) 法语专业；信息工程 (IE) 辅修 中国上海.

科研与实习经历

科研实习生 Mar–Jul 2026	智元机器人 - 参与 Longcheer 手机包装任务的强化学习后训练。 - 研究轮足机器人全身控制，包括腰部与底盘数据采集和相对位控，以提升落脚精度。
科研实习生 Jun–Sep 2024	阿里巴巴 / 阿里云 研究视觉导航与目标条件表征学习，重点探索 ViNT、NoMaD 等视觉导航方法。
产品实习生 Jun–Aug 2022	阿里云 Led product carbon-footprint certification for the Hangzhou Asian Games mascot plush toy, the Games-first zero-carbon licensed product, and supported the China Academy of Art low-carbon platform as a secondary project.

项目经历

工业项目

Main Contributor Mar–Jul 2026	基于实机强化学习后训练的 Longcheer 手机包装 Developed and benchmarked DAGger, AWR, and ALOE for real-robot RL post-training on Agibot's G02 platform. Achieved 90% success on the long-horizon Longcheer phone-packaging task, establishing a repeatable evaluation pipeline for mobile manipulation.
Main Contributor Jun–Aug 2022	Product Carbon Footprint and Low-Carbon Operations, 阿里云 结合生产过程数据及 CPCD-China、Ecoinvent 排放因子，构建产品全生命周期碳足迹评估流程。 完成原材料、生产和包装环节的碳排放核算，支持碳中和、第三方评估及数字化零碳标签，并交付杭州亚运会吉祥物认证报告。 为中国美术学院低碳平台定期交付校园低碳分析周报。

学术项目

Undergraduate Innovation Practice Program (IPP) Sep 2022–Jun 2024	Open-World Person Tracking and Identity Recognition Developed a working prototype for person detection, tracking, face recognition, and video retrieval in unconstrained surveillance scenes. Reproduced RetinaFace with ResNet-50 and YOLOv5 detection, and established initial gait-face alignment and multimodal-fusion baselines.
Participation in Research Program (PRP) Jul 2021–Jan 2022	Energy Internet Platform Data Interface Design and Development Developed data interfaces for an integrated energy demonstration system in a smart-grid building. Built Schneider Electric-based control and visualization modules that ingested and displayed real-time energy-operation data, delivering a working monitoring prototype.

论文发表

The ^{*} symbol denotes equal contribution, and the [†] symbol denotes corresponding authors. 会议论文

C1 Yijin Zhou, Linqian Zeng, Xiaoya Lu, Wenyuan Xie, Dongrui Liu, Junchi Yan, Jing Shao. “Exploring Agentic Tool-Calling Decisions via Uncertainty-Aligned Reinforcement Learning.” *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026. [arXiv:2606.06976](#).

C2 Guodong Zhang, Qichen He, Wenyuan Xie, Shaokai Wu, Yanbiao Ji, Qiuchang Li, Bayram Bayramli, Yue Ding, Hongtao Lu. “RelAfford6D: Relational 6D Affordance Graphs for Constraint-Driven Robotic Manipulation.” *European Conference on Computer Vision (ECCV)*, 2026. [arXiv:2606.27036](#).

C3 Yanbiao Ji, Qiuchang Li, Yuting Hu, Shaokai Wu, Wenyuan Xie, Guodong Zhang, Qicheng He, Deyi Ji, Yue Ding, Hongtao Lu. “Recovering Hidden Reward in Diffusion-Based Policies.” *International Conference on Machine Learning (ICML)*, 2026. [arXiv:2605.00623](#).

C4 Yijin Zhou, Yutang Ge, Wenyuan Xie, Linqian Zeng, Xiaowen Dong, Yuguang Wang. “How Out-of-Distribution Detection Learning Theory Enhances Transformer: Learnability and Reliability.” *International Conference on Machine Learning (ICML)*, 2026. [arXiv:2406.12915](#).

C5 Wenyuan Xie, Shaokai Wu, Yijin Zhou, Yanbiao Ji, Guodong Zhang, Bayram Bayramli, Qiuchang Li, Xunchu Zhou, Yue Ding, Hongtao Lu. “MVP-Nav: Multi-layer Value Map Planner Navigator.” *Robotics: Science and Systems (RSS)*, 2026. [arXiv:2606.31919](#).

C6 Shaokai Wu, Yanbiao Ji, Qiuchang Li, Zhiyi Zhang, Qichen He, Wenyuan Xie, Guodong Zhang, Bayram Bayramli, Yue Ding[†], Hongtao Lu[†]. “Dejavu: Towards Experience Feedback Learning for Embodied Intelligence.” *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [arXiv:2510.10181](#).

预印本

P1 Wenyuan Xie, Mengyang Hong, Yongzhong Wang, Yanbiao Ji, Yijin Zhou, Shaokai Wu, Shalayiding Sirejiding, Huayi Zhou, Yi-Chao Chen, Ma Ling, Yue Ding, Hongtao Lu. “Navi-Agent: Unlocalized Monocular Navigation Agent.” *arXiv preprint*, 2026. [arXiv:2609.20388](#).

P2 Yanbiao Ji, Qiuchang Li, Shaokai Wu, Wenyuan Xie, Deyi Ji, Qi Zhu, Yue Ding, Hongtao Lu. “Capability Loss in Embodied Recursive Self-Improvement.” *International Conference on Learning Representations (ICLR)*, 2027, under review. [OpenReview: IkYmTAyPXX](#).

获奖经历

SPEIT 优秀新生奖学金	2020–2021; CNY 10,000.
上海交通大学本科生学业优秀奖学金	2022–2023, 2023–2024; CNY 1,000 各项。
上海交通大学研究生学业奖学金	2024–2025; CNY 7,800.
上海交通大学篮球体总杯	2020–2021, 2022–2023; 八强。
1st Zhiyuan Cup Graduate Men’s 3-on-3 Basketball Tournament	2025; 第三名。

技能

编程	C++, Java, Python, PyTorch, Git.
语言	Chinese, English, French.